```
from google.colab import files

uploaded = files.upload()
```

Choose Files 3 files
airlines.csv(text/csv) - 359 bytes, last modified: 10/28/2026 - 100% done
flights.csv.zip(application/x-zip-compressed) - 200056978 bytes, last modified: 10/28/2026 - 100% done
airports.csv(text/csv) - 23867 bytes, last modified: 10/28/2026 - 100% done
Saving airlines.csv to airlines (2).csv
Saving flights.csv.zip to flights.csv.zip
Saving airports.csv to airports.csv

```
import os
os.listdir()
```

```
['.config',
 'airlines.csv',
 'airlines (1).csv',
 'airlines (2).csv',
 'airports.csv',
 'flights.csv.zip',
 'sample_data']
```

```
import pandas as pd

df = pd.read_csv("flights.csv.zip", nrows=10000)

df.head()
```

	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_DEPARTURE
0	2015	1	1	4	AS	98	N407AS	ANC	SEA	5
1	2015	1	1	4	AA	2336	N3KUAA	LAX	PBI	10
2	2015	1	1	4	US	840	N171US	SFO	CLT	20
3	2015	1	1	4	AA	258	N3HYAA	LAX	MIA	20
4	2015	1	1	4	AS	135	N527AS	SEA	ANC	25

5 rows × 31 columns

```
# Number of rows and columns
print("Dataset shape:", df.shape)

# Column names
print("\nColumn names:")
print(df.columns.tolist())

# Information about the dataset
print("\nDataset information:")
df.info()
```

```
Dataset shape: (10000, 31)

Column names:
['YEAR', 'MONTH', 'DAY', 'DAY_OF_WEEK', 'AIRLINE', 'FLIGHT_NUMBER', 'TAIL_NUMBER', 'ORIGIN_AIRPORT', 'DESTINATION_AIRPORT', 'SCH

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 31 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   YEAR                 10000 non-null  int64
 1   MONTH                10000 non-null  int64
 2   DAY                  10000 non-null  int64
 3   DAY_OF_WEEK          10000 non-null  int64
 4   AIRLINE              10000 non-null  object
 5   FLIGHT_NUMBER        10000 non-null  int64
 6   TAIL_NUMBER          9987 non-null   object
 7   ORIGIN_AIRPORT       10000 non-null  object
 8   DESTINATION_AIRPORT  10000 non-null  object
```

```
 9   SCHEDULED_DEPARTURE  10000 non-null  int64
 10  DEPARTURE_TIME       9612 non-null   float64
 11  DEPARTURE_DELAY      9612 non-null   float64
 12  TAXI_OUT             9611 non-null   float64
 13  WHEELS_OFF           9611 non-null   float64
 14  SCHEDULED_TIME       10000 non-null  int64
 15  ELAPSED_TIME         9593 non-null   float64
 16  AIR_TIME             9593 non-null   float64
 17  DISTANCE             10000 non-null  int64
 18  WHEELS_ON            9605 non-null   float64
 19  TAXI_IN              9605 non-null   float64
 20  SCHEDULED_ARRIVAL    10000 non-null  int64
 21  ARRIVAL_TIME         9605 non-null   float64
 22  ARRIVAL_DELAY        9593 non-null   float64
 23  DIVERTED             10000 non-null  int64
 24  CANCELLED            10000 non-null  int64
 25  CANCELLATION_REASON  392 non-null    object
 26  AIR_SYSTEM_DELAY     1728 non-null   float64
 27  SECURITY_DELAY       1728 non-null   float64
 28  AIRLINE_DELAY        1728 non-null   float64
 29  LATE_AIRCRAFT_DELAY  1728 non-null   float64
 30  WEATHER_DELAY        1728 non-null   float64
dtypes: float64(15), int64(11), object(5)
memory usage: 2.4+ MB
```

```
print("Dataset shape:", df.shape)
df.info()
```

```
Dataset shape: (10000, 31)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 31 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   YEAR                 10000 non-null  int64
 1   MONTH                10000 non-null  int64
 2   DAY                  10000 non-null  int64
 3   DAY_OF_WEEK          10000 non-null  int64
 4   AIRLINE              10000 non-null  object
 5   FLIGHT_NUMBER        10000 non-null  int64
 6   TAIL_NUMBER          9987 non-null   object
 7   ORIGIN_AIRPORT       10000 non-null  object
 8   DESTINATION_AIRPORT  10000 non-null  object
 9   SCHEDULED_DEPARTURE  10000 non-null  int64
 10  DEPARTURE_TIME       9612 non-null   float64
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 12  TAXI_OUT             9611 non-null   float64
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 15  ELAPSED_TIME         9593 non-null   float64
 16  AIR_TIME             9593 non-null   float64
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 20  SCHEDULED_ARRIVAL    10000 non-null  int64
 21  ARRIVAL_TIME         9605 non-null   float64
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 23  DIVERTED             10000 non-null  int64
 24  CANCELLED            10000 non-null  int64
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 29  LATE_AIRCRAFT_DELAY  1728 non-null   float64
 30  WEATHER_DELAY        1728 non-null   float64
dtypes: float64(15), int64(11), object(5)
memory usage: 2.4+ MB
```

```
print(df.isnull().sum())
```

```
YEAR                     0
MONTH                    0
DAY                      0
DAY_OF_WEEK              0
AIRLINE                  0
FLIGHT_NUMBER            0
TAIL_NUMBER             13
ORIGIN_AIRPORT           0
DESTINATION_AIRPORT      0
SCHEDULED_DEPARTURE      0
DEPARTURE_TIME         388
DEPARTURE_DELAY        388
```

```
TAXI_OUT               389
WHEELS_OFF             389
SCHEDULED_TIME           0
ELAPSED_TIME           407
AIR_TIME               407
DISTANCE                 0
WHEELS_ON              395
TAXI_IN                395
SCHEDULED_ARRIVAL        0
ARRIVAL_TIME           395
ARRIVAL_DELAY          407
DIVERTED                 0
CANCELLED                0
CANCELLATION_REASON    9608
AIR_SYSTEM_DELAY       8272
SECURITY_DELAY         8272
AIRLINE_DELAY          8272
LATE_AIRCRAFT_DELAY    8272
WEATHER_DELAY          8272
dtype: int64
```

```
print("Duplicate rows:", df.duplicated().sum())
```

```
Duplicate rows: 0
```

```
df = df.drop_duplicates()
print("Shape after removing duplicates:", df.shape)
```

```
Shape after removing duplicates: (10000, 31)
```

```
empty_columns = df.columns[df.isnull().all()].tolist()
print("Completely empty columns:", empty_columns)

df = df.dropna(axis=1, how="all")
```

```
Completely empty columns: []
```

```
for col in df.columns:
    if df[col].isnull().any():
        if df[col].dtype in ["int64", "float64"]:
            df[col] = df[col].fillna(df[col].median())
        else:
            mode_values = df[col].mode()
            if not mode_values.empty:
                df[col] = df[col].fillna(mode_values[0])

print("Missing values remaining:", df.isnull().sum().sum())
```

```
Missing values remaining: 0
```

```
print(df.dtypes)
```

```
YEAR                    int64
MONTH                   int64
DAY                     int64
DAY_OF_WEEK             int64
AIRLINE                object
FLIGHT_NUMBER           int64
TAIL_NUMBER            object
ORIGIN_AIRPORT         object
DESTINATION_AIRPORT    object
SCHEDULED_DEPARTURE     int64
DEPARTURE_TIME        float64
DEPARTURE_DELAY       float64
TAXI_OUT              float64
WHEELS_OFF            float64
SCHEDULED_TIME          int64
ELAPSED_TIME          float64
AIR_TIME              float64
DISTANCE                int64
WHEELS_ON             float64
TAXI_IN               float64
SCHEDULED_ARRIVAL       int64
ARRIVAL_TIME          float64
ARRIVAL_DELAY         float64
DIVERTED                int64
CANCELLED               int64
CANCELLATION_REASON    object
AIR_SYSTEM_DELAY      float64
```

```
SECURITY_DELAY         float64
AIRLINE_DELAY          float64
LATE_AIRCRAFT_DELAY    float64
WEATHER_DELAY          float64
dtype: object
```

```
print("Final shape:", df.shape)
print("Total missing values:", df.isnull().sum().sum())
print("Total duplicate rows:", df.duplicated().sum())

display(df.head())
```

```
Final shape: (10000, 31)
Total missing values: 0
Total duplicate rows: 0
```

1 to 5 of 5 entries Filter

index	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER	ORIGIN_AIRPORT	DESTINATION_AIRPORT	SCHEDULED_I
0	2015	1	1	4	AS	98	N407AS	ANC	SEA	
1	2015	1	1	4	AA	2336	N3KUAA	LAX	PBI	
2	2015	1	1	4	US	840	N171US	SFO	CLT	
3	2015	1	1	4	AA	258	N3HYAA	LAX	MIA	
4	2015	1	1	4	AS	135	N527AS	SEA	ANC	

Show 25 per page
Like what you see? Visit the data table notebook to learn more about interactive tables.
Warning: Total number of columns (31) exceeds max columns (20) limiting to first (20) columns.

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

# 1. Distribution of arrival delays
plt.figure(figsize=(8, 5))
sns.histplot(df["ARRIVAL_DELAY"], bins=50, kde=True)
plt.title("Distribution of Flight Arrival Delays")
plt.xlabel("Arrival Delay (minutes)")
plt.ylabel("Number of Flights")
plt.xlim(-60, 180)
plt.show()

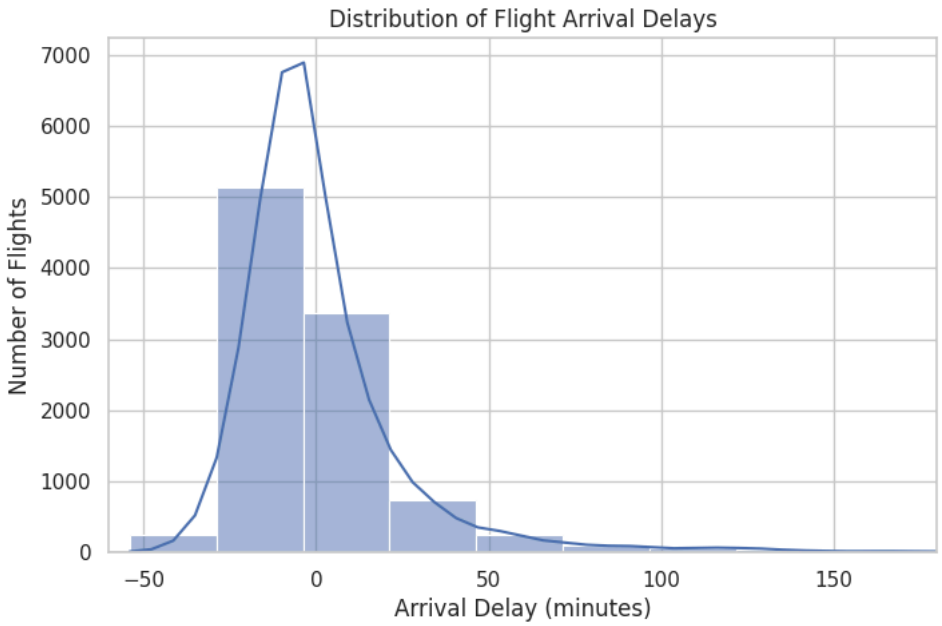
# 2. Cancelled vs non-cancelled flights
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="CANCELLED")
plt.title("Cancelled vs Non-Cancelled Flights")
plt.xlabel("Cancelled (0 = No, 1 = Yes)")
plt.ylabel("Number of Flights")
plt.show()

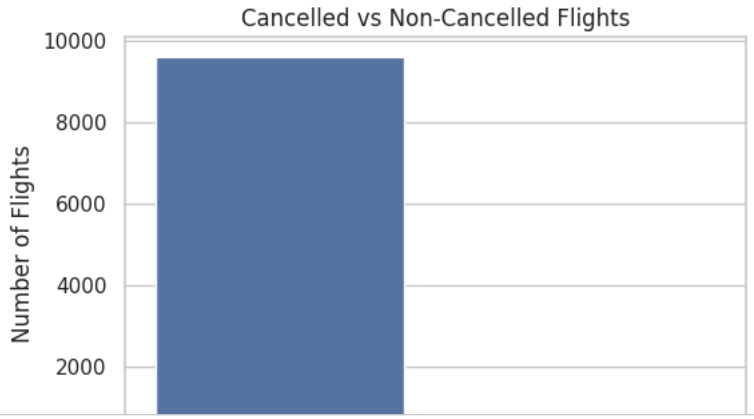
# 3. Distribution of departure delays
plt.figure(figsize=(8, 5))
sns.histplot(df["DEPARTURE_DELAY"], bins=50, kde=True)
plt.title("Distribution of Flight Departure Delays")
plt.xlabel("Departure Delay (minutes)")
plt.ylabel("Number of Flights")
plt.xlim(-60, 180)
plt.show()

# 4. Average arrival delay by airline
plt.figure(figsize=(10, 5))
sns.barplot(data=df, x="AIRLINE", y="ARRIVAL_DELAY",
            errorbar=None)
plt.title("Average Arrival Delay by Airline")
plt.xlabel("Airline")
plt.ylabel("Average Arrival Delay (minutes)")
plt.xticks(rotation=45)
plt.show()

# 5. Flight diversion counts
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="DIVERTED")
plt.title("Diverted vs Non-Diverted Flights")
plt.xlabel("Diverted (0 = No, 1 = Yes)")
plt.ylabel("Number of Flights")
```

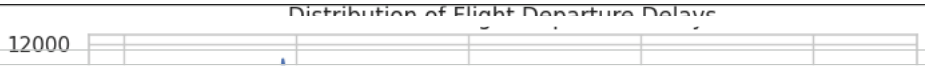
```
plt.show()
```






```
df.to_csv("cleaned_flight_data.csv", index=False)

from google.colab import files
files.download("cleaned_flight_data.csv")
```




```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

# Visualization 1: Arrival Delay Distribution
plt.figure(figsize=(8, 5))
sns.histplot(data=df, x="ARRIVAL_DELAY", bins=40)
plt.title("Distribution of Flight Arrival Delays")
plt.xlabel("Arrival Delay (minutes)")
plt.ylabel("Number of Flights")
plt.xlim(-60, 180)
plt.show()

# Visualization 2: Cancelled Flights
plt.figure(figsize=(6, 4))
sns.countplot(data=df, x="CANCELLED")
plt.title("Cancelled vs Non-Cancelled Flights")
plt.xlabel("Cancelled (0 = No, 1 = Yes)")
plt.ylabel("Number of Flights")
plt.show()

# Visualization 3: Average Arrival Delay by Airline
airline_delay = (
    df.groupby("AIRLINE")["ARRIVAL_DELAY"]
    .mean()
```

```
    .sort_values(ascending=False)
    .reset_index()
)

plt.figure(figsize=(9, 5))
sns.barplot(data=airline_delay, x="AIRLINE", y="ARRIVAL_DELAY")
plt.title("Average Arrival Delay by Airline")
plt.xlabel("Airline")
plt.ylabel("Average Arrival Delay (minutes)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

